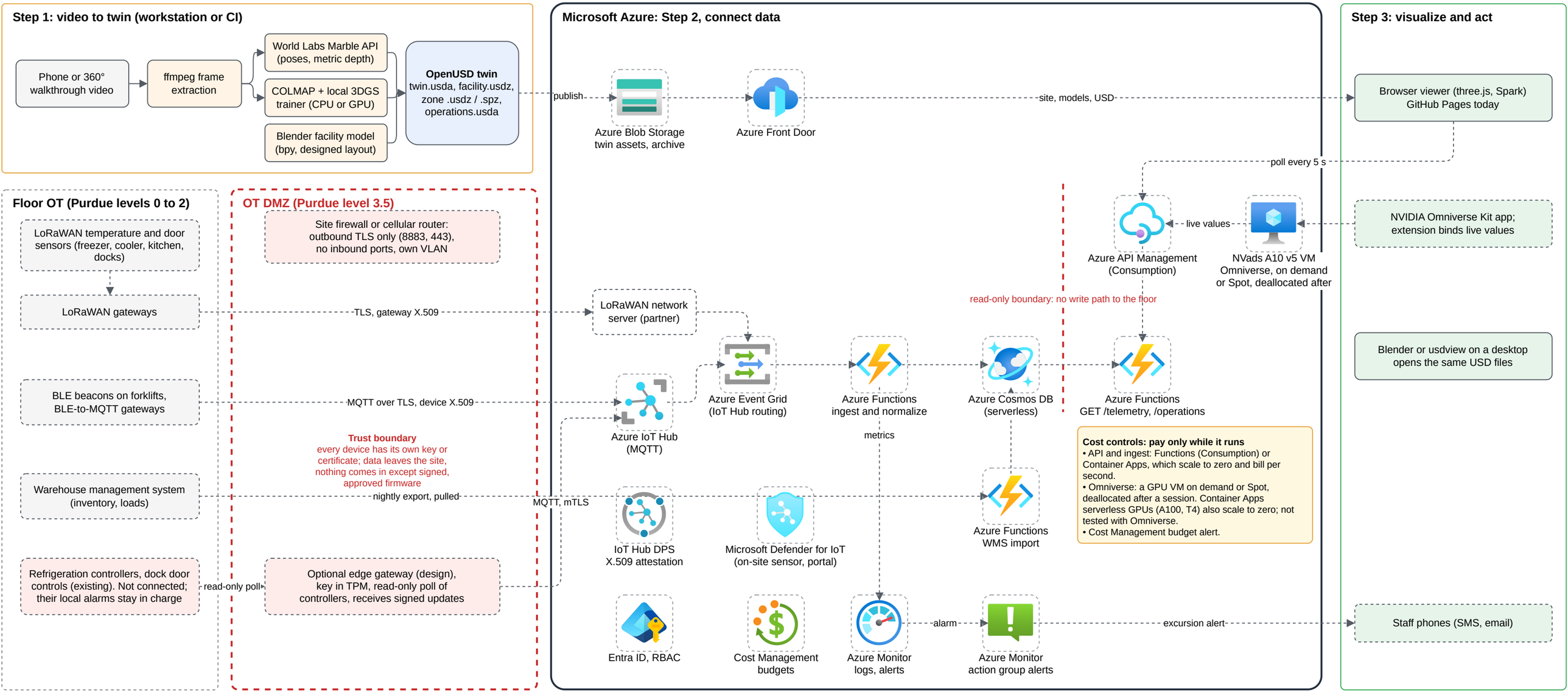
Food bank digital twin on Microsoft Azure



OT security: zones, conduits and what is built (docs/OT_SECURITY.md)
Field devices and edge gateways sit on the floor; the OT DMZ passes only outbound TLS; cloud ingest accepts data and sends no commands; the twin is read-only. Existing refrigeration and dock controls are not connected. Red dashed lines are trust boundaries. The only designed path toward the floor is a signed, human-approved firmware update.
Built: nothing on Azure; this is a design only.
Design only (dashed): per-device certificates and keys, least-privilege device policies, segmentation, the edge gateway, device monitoring, plausibility and heartbeat checks on readings, signed updates.

Equivalent design, not deployed. Azure IoT Hub has no built-in LoRaWAN network server, so a partner network server forwards uplinks to IoT Hub.